

using a Bland-Altman¹⁰ plot of all samples with values in both assays (35 samples). The overall bias between the laboratory-developed assay and ASR was calculated from this plot using GraphPad Prism 5 Software (GraphPad Software, San Diego, CA). In addition, using proficiency samples provided by the American Society of Transplantation (AST, Mt Laurel, NJ) and Quality Control for Molecular Diagnostics (QCMD, Glasgow, Scotland), we determined results obtained from both assays.

Results

Limits of Detection and Quantification

The LOD of the ASR was 2.25 log₁₀ copies/mL (180 copies/mL) (Table 1), which was equivalent to 1.30 log₁₀ copies/mL (20 copies/mL) using the Namalwa cell line. The LOQ based on an SD of 0.15 log₁₀ copies/mL or less was 3.90 log₁₀ copies/mL (Table 1). Of note, the SDs at 3.53 log₁₀ copies/mL and 2.89 log₁₀ copies/mL were 0.18 and 0.17 log₁₀ copies/mL, respectively.

Linearity and Reproducibility

The assay was linear from 2.17 to 6.21 log₁₀ copies/mL (Figure 1), with an R^2 value of 0.9925 ($y = 1.0119x - 0.1184$). Reproducibility was assessed by testing 7 replicates of low and high viral load patient samples and 6 replicates of the Qs. The low patient sample had a mean (SD, CV) value of 3.70 log₁₀ copies/mL (0.07, 2.0%). The high patient sample had a mean (SD, CV) value of 5.40 log₁₀ copies/mL (0.07, 1.4%). The Qs had mean Ct (SD, CV) values as follows: QS1, 21.8 (0.4, 1.6%); QS2, 25.1 (0.4, 1.5%); QS3, 28.5 (0.2, 0.8%); and QS4, 31.8 (0.3, 1.0%).

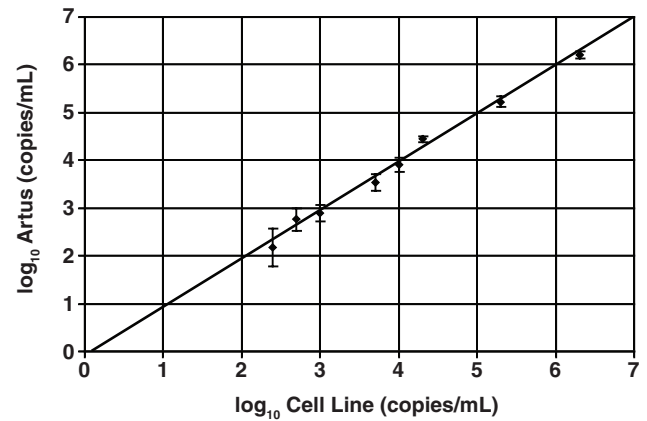


Figure 1 Linearity plot. The mean values for each concentration from 2.44 to 6.21 log₁₀ copies/mL were plotted, with error bars representing the SD. A linear regression line ($y = 1.0119x - 0.1184$) was plotted with Microsoft Excel 2003 software (Redmond, WA), with an R^2 value of 0.9925, indicating that the assay is linear at least across this range.

Agreement

Of the 61 patient samples tested by the laboratory-developed method and the ASR, 35 were positive in both assays. Of the 6 samples that were positive in the Artus assay but not in the laboratory-developed assay, the average copy number was 1.62 log₁₀ copies/mL, with a range of 1.41 to 2.13 log₁₀ copies/mL. There was only 1 sample that was positive in the laboratory-developed assay but negative in the Artus assay, with a value of 3.23 log₁₀ copies/mL. The remaining 19 samples were negative by both assays. The 35 samples that were positive in both assays were plotted in a standard Bland-Altman¹⁰ plot (Figure 2). The mean difference between

Table 1
Dilution Series*

EBV Cell Line (log ₁₀ copies/mL)	Mean Artus (log ₁₀ copies/mL)	log ₁₀ Artus SD	log ₁₀ Artus CV	No. Tested/No. Positive (% Positive)
6.30	6.21	0.07	0.01	8/8 (100)
5.30	5.23	0.11	0.02	8/8 (100)
4.30	4.44	0.07	0.01	8/8 (100)
4.00	3.90	0.15	0.04	12/12 (100)
3.70	3.53	0.18	0.05	12/12 (100)
3.00	2.89	0.17	0.06	12/12 (100)
2.70	2.76	0.24	0.09	16/16 (100)
2.40	2.17	0.39	0.02	16/16 (100)
2.00	2.44	0.33	0.14	8/8 (100)
1.30	2.25	0.48	0.21	8/8 (100)
0.60	1.60	1.06	0.66	6/8 (75)
-0.10	1.17	1.02	0.87	5/8 (63)

CV, coefficient of variation; EBV, Epstein-Barr virus.

* The expected copy number based on the approximate initial number of Namalwa cells (ATCC CRL-1432), the mean measured copy number using the Artus assay, SD, CV, and the number and percentage of positive replicates for each dilution of the Namalwa cell line in EBV- whole blood were tabulated.